

Program Analysis 2025-26

Lecture #21

Control Flow Analysis



UNIVERSITÀ
DI PISA

Abstract Values as Complete Lattices

- Instead of using sets of concrete data, we can use structured abstract domains
- We separate:
 - **Term** information
 - **Data** information

Two components

$$\hat{v} \in \widehat{\mathbf{Val}}_d = \wp(\mathbf{Term} \cup \mathbf{Data}) \equiv \wp(\mathbf{Term}) \times \wp(\mathbf{Data})$$

- replace $\wp(\mathbf{Data})$ by a **complete lattice** L
- and perform a development closely related to that of the **(forward) Monotone Frameworks**

Abstract Values as Complete Lattices

A **monotone structure** consists of:

- a complete lattice L , and
- a set F of monotone functions of $L \times L \rightarrow L$

An **instance** of a monotone structure then consists of the structure (L, F)

- a mapping ι from the constants $c \in \mathbf{Const}$ to values in L , and
- a mapping f from the binary operators $op \in \mathbf{Op}$ to functions of F

No need of flow component: it is up to the Control Flow Analysis to determine the flow

Example

Lift concrete operations to abstract values

Take the following **monotone structure**:

- $L = \wp(\mathbf{Data})$
- F : monotone functions of $L \times L \rightarrow L$

An **instance** of this monotone structure is obtained by taking

- $\iota_c = \{d_c\}$ for each constant $c \in \mathbf{Const}$ (and with $d_c \in \mathbf{Data}$)
- $f_{op}(l_1, l_2) = \bigcup \{d_{op}(d_1, d_2) \mid d_1 \in l_1, d_2 \in l_2\}$

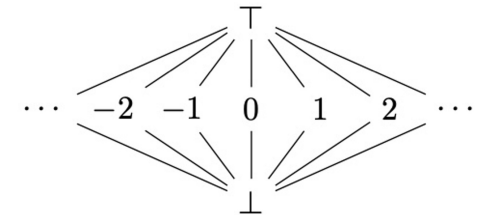
for all $op \in \mathbf{Op}$ (and where $d_{op} : \mathbf{Data} \times \mathbf{Data} \rightarrow \mathbf{Data}$ is like before)

Apply the
operator to all
pairs of possible
values

Example: Constant Propagation Analysis

A **monotone structure** for Constant Propagation Analysis

- $L = \mathbb{Z}_{\perp}^{\top} \times \wp(\{\text{tt}, \text{ff}\})$ with $\mathbb{Z}_{\perp}^{\top}$ is *flat*($\mathbb{Z}$) where $\top$ indicates that a variable is non constant
- F : monotone functions of $L \times L \rightarrow L$



An **instance** of this monotone structure is obtained by taking, e.g.,

$$l_c = (7, \emptyset) \text{ and } l_{\text{true}} = (\perp, \{\text{tt}\})$$

$$f_+(l_1, l_2) = \begin{cases} (z_1 + z_2, \emptyset) & \text{if } l_1 = (z_1, \dots), l_2 = (z_2, \dots) \text{ and } z_1, z_2 \in \mathbb{Z} \\ (\perp, \emptyset) & \text{if } l_1 = (z_1, \dots), l_2 = (z_2, \dots) \text{ and } z_1 = \perp \text{ or } z_2 = \perp \in \mathbb{Z} \\ (\top, \emptyset) & \text{otherwise} \end{cases}$$

For example:

$$f_+(l_c, l_{\text{true}}) = (\perp, \emptyset) \text{ since } z_1 \in \mathbb{Z} \text{ and } z_2 = \perp$$

f_+ propagates constants when possible, otherwise returns $\perp$ (unknown)

New CFA

We combine CFA and DFA information, by tracking both functions and data simultaneously

Abstract domains

$$\widehat{\mathbf{Val}}_d = \wp(\mathbf{Term}) \times \wp(\mathbf{Data})$$

decoupling

CFA part

DFA part

$$\begin{aligned} \hat{v} &\in \widehat{\mathbf{Val}} = \wp(\mathbf{Term}) \\ \hat{\rho} &\in \widehat{\mathbf{Env}} : \mathbf{Var} \rightarrow \widehat{\mathbf{Val}} \\ \hat{C} &\in \widehat{\mathbf{Cache}} : \mathbf{Lab} \rightarrow \widehat{\mathbf{Val}} \end{aligned}$$

$$\begin{aligned} \hat{d} &\in \widehat{\mathbf{Data}} = L \\ \hat{\delta} &\in \widehat{\mathbf{DEnv}} : \mathbf{Var} \rightarrow \widehat{\mathbf{Data}} \\ \hat{D} &\in \widehat{\mathbf{DCache}} : \mathbf{Lab} \rightarrow \widehat{\mathbf{Data}} \end{aligned}$$

Control-flow and data-flow are computed together

Acceptability relation

$$(\hat{C}, \hat{D}, \hat{\rho}, \hat{\delta}) \models_D e$$

Clause for constants

$$[con] (\hat{C}, \hat{D}, \hat{\rho}, \hat{\delta}) \models_D c^\ell \text{ iff } \iota_c \sqsubseteq \hat{D}(\ell)$$

At label ℓ , the abstract data value of c must be recorded
 ι tells us which abstract value a constant contributes

Constants inject
data-flow information
into the analysis

Example

$$c = 7$$

$$\iota_c = \{ + \}$$

$$\Rightarrow \hat{D}(\ell) \text{ must contain } \{ + \}$$

Clause for binary operation

$$[op] (\hat{C}, \hat{D}, \hat{\rho}, \hat{\delta}) \models_D (t_1^{\ell_1} \text{ op } t_2^{\ell_2})^\ell \text{ iff } (\hat{C}, \hat{D}, \hat{\rho}, \hat{\delta}) \models_D t_1^{\ell_1} \wedge (\hat{C}, \hat{D}, \hat{\rho}, \hat{\delta}) \models_D t_2^{\ell_2} \wedge$$
$$f_{op}(\hat{D}(\ell_1), \hat{D}(\ell_2)) \sqsubseteq \hat{D}(\ell)$$

Now, we use the component f .

Compute the abstract result of the operation and propagate it to the result label

Example

$x \in \{+, 0\}, y \in \{+\}$

$\Rightarrow x + y \in \{+, 0, -\}$

$\Rightarrow \hat{D}(\ell)$ must include $\{+, 0, -\}$

Clause for conditional

$$[if] (\hat{C}, \hat{D}, \hat{\rho}, \hat{\delta}) \models_D (\text{if } t_0^{\ell_0} \text{ then } t_1^{\ell_1} \text{ else } t_2^{\ell_2})^\ell$$

$$\text{iff } (\hat{C}, \hat{D}, \hat{\rho}, \hat{\delta}) \models_D t_0^{\ell_0} \wedge$$

$$(\iota_{\text{true}} \sqsubseteq \hat{D}(\ell_0) \Rightarrow (\hat{C}, \hat{D}, \hat{\rho}, \hat{\delta}) \models_D t_1^{\ell_1} \wedge \hat{C}(\ell_1) \subseteq \hat{C}(\ell) \wedge \hat{D}(\ell_1) \sqsubseteq \hat{D}(\ell)) \wedge$$

$$(\iota_{\text{false}} \sqsubseteq \hat{D}(\ell_0) \Rightarrow (\hat{C}, \hat{D}, \hat{\rho}, \hat{\delta}) \models_D t_2^{\ell_2} \wedge \hat{C}(\ell_2) \subseteq \hat{C}(\ell) \wedge \hat{D}(\ell_2) \sqsubseteq \hat{D}(\ell))$$

The clause [if]

- analyze condition
- use abstract boolean value and propagate only analysed branches

Example

$$e \triangleq (\text{if } (3^1 > 0^2)^3 \text{ then } (\text{fn } y \Rightarrow y^4)^5 \text{ else } (\text{fn } z \Rightarrow z^6)^7)^8$$

$$\iota_{\text{true}} \sqsubseteq \hat{D}(3) \Rightarrow \hat{C}(5) \subseteq \hat{C}(8) \text{ and } \hat{D}(5) \subseteq \hat{D}(8) \dots \iota_{\text{false}} \not\sqsubseteq \hat{D}(3) \Rightarrow$$

CFA with abstract values as complete lattices

$$\begin{aligned}
 [\text{con}] \quad & (\widehat{C}, \widehat{D}, \widehat{\rho}, \widehat{\delta}) \models_D c^\ell \text{ iff } \iota_c \subseteq \widehat{D}(\ell) \\
 [\text{var}] \quad & (\widehat{C}, \widehat{D}, \widehat{\rho}, \widehat{\delta}) \models_D x^\ell \text{ iff } \widehat{\rho}(x) \subseteq \widehat{C}(\ell) \wedge \widehat{\delta}(x) \subseteq \widehat{D}(\ell) \\
 [\text{fn}] \quad & (\widehat{C}, \widehat{D}, \widehat{\rho}, \widehat{\delta}) \models_D (\text{fn } x \Rightarrow e_0)^\ell \text{ iff } \{\text{fn } x \Rightarrow e_0\} \subseteq \widehat{C}(\ell) \\
 [\text{fun}] \quad & (\widehat{C}, \widehat{D}, \widehat{\rho}, \widehat{\delta}) \models_D (\text{fun } f \ x \Rightarrow e_0)^\ell \text{ iff } \{\text{fun } f \ x \Rightarrow e_0\} \subseteq \widehat{C}(\ell) \\
 [\text{app}] \quad & (\widehat{C}, \widehat{D}, \widehat{\rho}, \widehat{\delta}) \models_D (t_1^{\ell_1} \ t_2^{\ell_2})^\ell \\
 & \text{ iff } (\widehat{C}, \widehat{D}, \widehat{\rho}, \widehat{\delta}) \models_D t_1^{\ell_1} \wedge (\widehat{C}, \widehat{D}, \widehat{\rho}, \widehat{\delta}) \models_D t_2^{\ell_2} \wedge \\
 & \quad (\forall (\text{fn } x \Rightarrow t_0^{\ell_0}) \in \widehat{C}(\ell_1) : \\
 & \quad \quad (\widehat{C}, \widehat{D}, \widehat{\rho}, \widehat{\delta}) \models_D t_0^{\ell_0} \wedge \\
 & \quad \quad \widehat{C}(\ell_2) \subseteq \widehat{\rho}(x) \wedge \widehat{D}(\ell_2) \subseteq \widehat{\delta}(x) \wedge \\
 & \quad \quad \widehat{C}(\ell_0) \subseteq \widehat{C}(\ell) \wedge \widehat{D}(\ell_0) \subseteq \widehat{D}(\ell)) \wedge \\
 & \quad (\forall (\text{fun } f \ x \Rightarrow t_0^{\ell_0}) \in \widehat{C}(\ell_1) : \\
 & \quad \quad (\widehat{C}, \widehat{D}, \widehat{\rho}, \widehat{\delta}) \models_D t_0^{\ell_0} \wedge \\
 & \quad \quad \widehat{C}(\ell_2) \subseteq \widehat{\rho}(x) \wedge \widehat{D}(\ell_2) \subseteq \widehat{\delta}(x) \wedge \\
 & \quad \quad \widehat{C}(\ell_0) \subseteq \widehat{C}(\ell) \wedge \widehat{D}(\ell_0) \subseteq \widehat{D}(\ell) \wedge \\
 & \quad \quad \{\text{fun } f \ x \Rightarrow t_0^{\ell_0}\} \subseteq \widehat{\rho}(f)) \\
 [\text{if}] \quad & (\widehat{C}, \widehat{D}, \widehat{\rho}, \widehat{\delta}) \models_D (\text{if } t_0^{\ell_0} \text{ then } t_1^{\ell_1} \text{ else } t_2^{\ell_2})^\ell \\
 & \text{ iff } (\widehat{C}, \widehat{D}, \widehat{\rho}, \widehat{\delta}) \models_D t_0^{\ell_0} \wedge \\
 & \quad (\iota_{\text{true}} \subseteq \widehat{D}(\ell_0) \Rightarrow (\widehat{C}, \widehat{D}, \widehat{\rho}, \widehat{\delta}) \models_D t_1^{\ell_1} \wedge \\
 & \quad \quad \widehat{C}(\ell_1) \subseteq \widehat{C}(\ell) \wedge \\
 & \quad \quad \widehat{D}(\ell_1) \subseteq \widehat{D}(\ell)) \wedge \\
 & \quad (\iota_{\text{false}} \subseteq \widehat{D}(\ell_0) \Rightarrow (\widehat{C}, \widehat{D}, \widehat{\rho}, \widehat{\delta}) \models_D t_2^{\ell_2} \wedge \\
 & \quad \quad \widehat{C}(\ell_2) \subseteq \widehat{C}(\ell) \wedge \\
 & \quad \quad \widehat{D}(\ell_2) \subseteq \widehat{D}(\ell)) \\
 [\text{let}] \quad & (\widehat{C}, \widehat{D}, \widehat{\rho}, \widehat{\delta}) \models_D (\text{let } x = t_1^{\ell_1} \text{ in } t_2^{\ell_2})^\ell \\
 & \text{ iff } (\widehat{C}, \widehat{D}, \widehat{\rho}, \widehat{\delta}) \models_D t_1^{\ell_1} \wedge (\widehat{C}, \widehat{D}, \widehat{\rho}, \widehat{\delta}) \models_D t_2^{\ell_2} \wedge \\
 & \quad \widehat{C}(\ell_1) \subseteq \widehat{\rho}(x) \wedge \widehat{D}(\ell_1) \subseteq \widehat{\delta}(x) \wedge \\
 & \quad \widehat{C}(\ell_2) \subseteq \widehat{C}(\ell) \wedge \widehat{D}(\ell_2) \subseteq \widehat{D}(\ell) \\
 [\text{op}] \quad & (\widehat{C}, \widehat{D}, \widehat{\rho}, \widehat{\delta}) \models_D (t_1^{\ell_1} \text{ op } t_2^{\ell_2})^\ell \\
 & \text{ iff } (\widehat{C}, \widehat{D}, \widehat{\rho}, \widehat{\delta}) \models_D t_1^{\ell_1} \wedge (\widehat{C}, \widehat{D}, \widehat{\rho}, \widehat{\delta}) \models_D t_2^{\ell_2} \wedge \\
 & \quad f_{\text{op}}(\widehat{D}(\ell_1), \widehat{D}(\ell_2)) \subseteq \widehat{D}(\ell)
 \end{aligned}$$

Back to our Example 3

```
e  $\triangleq$  (let f = (fn x => (if (x1 > 02)3 then (fn y => y4)5
                        else (fn z => 256)7)8)9
    in ((f10 311)12 013)14)15
```

The last column describes the solution $(\hat{C}, \hat{D}, \hat{\rho}, \hat{\delta})$, where it is evident the splitting between

- the control flow component $(\hat{C}, \hat{\rho})$
- and the data flow component $(\hat{D}, \hat{\delta})$

Note, e.g., that

- x is always positive: $\delta(\hat{x}) = \{ + \}$
- condition is always true: $\hat{D}(3) = \{ \text{tt} \}$

	Pure CFA	CFA (powersets)	CFA (complete lattices)	
	Section 3.1	Subsection 3.5.1	Subsection 3.5.2	
	$(\widehat{C}, \widehat{\rho})$	$(\widehat{C}, \widehat{\rho})$	$(\widehat{C}, \widehat{\rho})$	$(\widehat{D}, \widehat{\delta})$
1	$\emptyset$	$\{+\}$	$\emptyset$	$\{+\}$
2	$\emptyset$	$\{0\}$	$\emptyset$	$\{0\}$
3	$\emptyset$	$\{tt\}$	$\emptyset$	$\{tt\}$
4	$\emptyset$	$\{0\}$	$\emptyset$	$\{0\}$
5	$\{fn\ y \Rightarrow y^4\}$	$\{fn\ y \Rightarrow y^4\}$	$\{fn\ y \Rightarrow y^4\}$	$\emptyset$
6	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$
7	$\{fn\ z \Rightarrow 25^6\}$	$\emptyset$	$\emptyset$	$\emptyset$
8	$\{fn\ y \Rightarrow y^4, fn\ z \Rightarrow 25^6\}$	$\{fn\ y \Rightarrow y^4\}$	$\{fn\ y \Rightarrow y^4\}$	$\emptyset$
9	$\{fn\ x \Rightarrow (\dots)^8\}$	$\{fn\ x \Rightarrow (\dots)^8\}$	$\{fn\ x \Rightarrow (\dots)^8\}$	$\emptyset$
10	$\{fn\ x \Rightarrow (\dots)^8\}$	$\{fn\ x \Rightarrow (\dots)^8\}$	$\{fn\ x \Rightarrow (\dots)^8\}$	$\emptyset$
11	$\emptyset$	$\{+\}$	$\emptyset$	$\{+\}$
12	$\{fn\ y \Rightarrow y^4, fn\ z \Rightarrow 25^6\}$	$\{fn\ y \Rightarrow y^4\}$	$\{fn\ y \Rightarrow y^4\}$	$\emptyset$
13	$\emptyset$	$\{0\}$	$\emptyset$	$\{0\}$
14	$\emptyset$	$\{0\}$	$\emptyset$	$\{0\}$
15	$\emptyset$	$\{0\}$	$\emptyset$	$\{0\}$
f	$\{fn\ x \Rightarrow (\dots)^8\}$	$\{fn\ x \Rightarrow (\dots)^8\}$	$\{fn\ x \Rightarrow (\dots)^8\}$	$\emptyset$
x	$\emptyset$	$\{+\}$	$\emptyset$	$\{+\}$
y	$\emptyset$	$\{0\}$	$\emptyset$	$\{0\}$
z	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$

Correctness of the analysis

The combined CFA + DFA is sound w.r.t. the operational semantics

- Same proof techniques as before apply
- The algorithm terminates if L satisfies the **Ascending Chain Condition (ACC)**, as in the case for Monotone Frameworks

no infinite strictly increasing sequences
⇒ fixpoint computation terminates



Staging the specification

- If we change the [if] clause in such a way that the data flow component cannot influence the control flow component, with a modified relation $\models'_D$, then $(\hat{C}, \hat{D}, \hat{\rho}, \hat{\delta}) \models'_D e$ guarantees that $(\hat{C}, \hat{\rho}) \models e$

We can solve:

1. control-flow ($\hat{C}$ and $\hat{\rho}$) first
2. data-flow ($\hat{D}$ and $\hat{\delta}$) afterwards

- The analysis still yields the **least solution** of the combined set of constraints

Back to Example 3

$$e \triangleq (\text{let } f = (\text{fn } x \Rightarrow (\text{if } (x^1 > 0^2)^3 \text{ then } (\text{fn } y \Rightarrow y^4)^5 \\ \text{else } (\text{fn } z \Rightarrow 25^6)^7)^8)^9 \\ \text{in } ((f^{10} \ 3^{11})^{12} \ 0^{13})^{14})^{15}$$

If we modify the [if] clause as discussed above then $\hat{C}$ and $\hat{\rho}$ are as in pure CFA, while $\hat{D}$ and $\hat{\delta}$ associate slightly larger sets with some labels and variables

- $\hat{D}(6) = \{ + \}$
- $\hat{D}(14) = \{ 0, + \}$
- $\hat{D}(15) = \{ 0, + \}$
- $\hat{\delta}(z) = \{ 0 \}$

	Pure CFA	CFA (powersets)	CFA (complete lattices)	
	Section 3.1	Subsection 3.5.1	Subsection 3.5.2	
	$(\widehat{C}, \widehat{\rho})$	$(\widehat{C}, \widehat{\rho})$	$(\widehat{C}, \widehat{\rho})$	$(\widehat{D}, \widehat{\delta})$
1	$\emptyset$	$\{+\}$	$\emptyset$	$\{+\}$
2	$\emptyset$	$\{0\}$	$\emptyset$	$\{0\}$
3	$\emptyset$	$\{tt\}$	$\emptyset$	$\{tt\}$
4	$\emptyset$	$\{0\}$	$\emptyset$	$\{0\}$
5	$\{fn\ y \Rightarrow y^4\}$	$\{fn\ y \Rightarrow y^4\}$	$\{fn\ y \Rightarrow y^4\}$	$\emptyset$
6	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$
7	$\{fn\ z \Rightarrow 25^6\}$	$\emptyset$	$\emptyset$	$\emptyset$
8	$\{fn\ y \Rightarrow y^4, \\ fn\ z \Rightarrow 25^6\}$	$\{fn\ y \Rightarrow y^4\}$	$\{fn\ y \Rightarrow y^4\}$	$\emptyset$
9	$\{fn\ x \Rightarrow (\dots)^8\}$	$\{fn\ x \Rightarrow (\dots)^8\}$	$\{fn\ x \Rightarrow (\dots)^8\}$	$\emptyset$
10	$\{fn\ x \Rightarrow (\dots)^8\}$	$\{fn\ x \Rightarrow (\dots)^8\}$	$\{fn\ x \Rightarrow (\dots)^8\}$	$\emptyset$
11	$\emptyset$	$\{+\}$	$\emptyset$	$\{+\}$
12	$\{fn\ y \Rightarrow y^4, \\ fn\ z \Rightarrow 25^6\}$	$\{fn\ y \Rightarrow y^4\}$	$\{fn\ y \Rightarrow y^4\}$	$\emptyset$
13	$\emptyset$	$\{0\}$	$\emptyset$	$\{0\}$
14	$\emptyset$	$\{0\}$	$\emptyset$	$\{0\}$
15	$\emptyset$	$\{0\}$	$\emptyset$	$\{0\}$
f	$\{fn\ x \Rightarrow (\dots)^8\}$	$\{fn\ x \Rightarrow (\dots)^8\}$	$\{fn\ x \Rightarrow (\dots)^8\}$	$\emptyset$
x	$\emptyset$	$\{+\}$	$\emptyset$	$\{+\}$
y	$\emptyset$	$\{0\}$	$\emptyset$	$\{0\}$
z	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$

More instances and precision

Example 4

$e \triangleq (\text{let } f = (\text{fn } x \Rightarrow x^1)^2$
 $\text{in } ((f^3 f^4)^5 (\text{fn } y \Rightarrow y^6)^7)^8)^9$

The least O-CFA analysis is given by $(\hat{C}_{id}, \hat{\rho}_{id})$

- $\hat{C}_{id}(1) = \{\text{fn } x \Rightarrow x^1, \text{fn } y \Rightarrow y^6\}$
- $\hat{C}_{id}(2) = \{\text{fn } x \Rightarrow x^1\}$
- $\hat{C}_{id}(3) = \{\text{fn } x \Rightarrow x^1\}$
- $\hat{C}_{id}(4) = \{\text{fn } x \Rightarrow x^1\}$
- $\hat{C}_{id}(5) = \{\text{fn } x \Rightarrow x^1, \text{fn } y \Rightarrow y^6\}$
- $\hat{C}_{id}(6) = \{\text{fn } y \Rightarrow y^6\}$
- $\hat{C}_{id}(7) = \{\text{fn } y \Rightarrow y^6\}$
- $\hat{C}_{id}(8) = \{\text{fn } x \Rightarrow x^1, \text{fn } y \Rightarrow y^6\}$
- $\hat{C}_{id}(9) = \{\text{fn } x \Rightarrow x^1, \text{fn } y \Rightarrow y^6\}$
- $\hat{\rho}_{id}(f) = \{\text{fn } x \Rightarrow x^1\}$
- $\hat{\rho}_{id}(x) = \{\text{fn } x \Rightarrow x^1, \text{fn } y \Rightarrow y^6\}$
- $\hat{\rho}_{id}(y) = \{\text{fn } y \Rightarrow y^6\}$

Only $\text{fn } y \Rightarrow y^6$ is a possible result of the evaluation of e
 O-CFA merges different calling contexts $\Rightarrow$ **loss of precision**

Example 4 bis

Idea: duplicate functions per call site

$e' \triangleq$ (let $f_1 = (\text{fn } x_1 \Rightarrow x_1^1)^2$
in (let $f_2 = (\text{fn } x_2 \Rightarrow x_2^3)^4$
in ((f_1^5 f_2^6)⁷ ($\text{fn } y \Rightarrow y^8$)⁹)¹⁰)¹¹)¹²

separate each use of f

• ...

• $\hat{C}_{id}(12) = \{\text{fn } y \Rightarrow y^8\}$

• $\hat{\rho}_{id}(x_1) = \{\text{fn } x_2 \Rightarrow x_2^3\}$

• $\hat{\rho}_{id}(x_2) = \{\text{fn } y \Rightarrow y^8\}$

Now each parameter is bound in a single context:

• x_1 can only be bound to $\text{fn } x_2 \Rightarrow x_2^3$ and

• x_2 can only be bound to $\text{fn } y \Rightarrow y^8$

• so no merging of flows $\Rightarrow$ precise result: $\text{fn } y \Rightarrow y^8$ only

precision comes from
separating flows

Adding Context Information

Uniform k -CFA Analysis

Adding contexts

- 0-CFA is a **context-insensitive** (or **mono-variant**) analysis
- It does not distinguish various instances of program variables and program points from each other, in particular does not distinguish different calling contexts
- **Context-sensitive** (or **poly-variant**) analysis does distinguish
 - Examples of such analyses include **k -CFA** analyses, uniform **k -CFA** analyses, polynomial **k -CFA** analyses and the Cartesian Product Algorithm
- We focus on **uniform k -CFA** (“uniform” because environment and cache have the same precision)

Abstract domains: contexts

Goal: distinguish different function calls

Idea: extend $\widehat{\mathbf{Val}}$ to include context information

- **Contexts** $\hat{\delta}$ record the k last dynamic call sites, e.g., for $k = 1$, the context amounts just to the label of the expression that called the current function
- In general, contexts are sequences of labels of length at most $\leq k$

$$\hat{\delta} \in \Delta = \mathbf{Lab}^{\leq k}$$

We track information separately for different call strings*

- Same function, different contexts $\Rightarrow$ different analyses

*A **call string** is a list of labels $\ell_1, \dots, \ell_n$ denoting that the last n calls were of the form $(e_{1n} \ e_{2n})^{\ell_n} \dots (e_{11} \ e_{21})^{\ell_1}$ with $(e_{11} \ e_{21})^{\ell_1}$ being the most recent call; we write ϵ for the empty call string

Abstract domains: context environment

- A **context environment** records in which context each variable was defined
- So we associate a context to each variable occurrence

Definition points of free
variables of terms

$$ce \in \mathbf{Cenv} = \mathbf{Var} \rightarrow \Delta$$

- The same variable may have different values in different contexts
- The context environment will play a role similar to the environment of the semantics

Context-sensitive abstract values

In our set of values, instead of just the function abstractions, we need to track **closures**, i.e., a pair of a function label and a context environment

- A value is paired with the context environment where it was created
- We shall extend the abstract values as

$$\hat{v} \in \widehat{\mathbf{Val}} = \wp(\mathbf{Val} \times \mathbf{Cenv})$$

Context-sensitive environments

- Abstract environment $\hat{\rho}$ and abstract cache $\hat{C}$ will now map a variable (a label, resp.) and a context to an abstract value

$$\begin{aligned}\hat{\rho} &\in \widehat{\mathbf{Cenv}} = (\mathbf{Var} \times \Delta) \rightarrow \widehat{\mathbf{Val}} \\ \hat{C} &\in \widehat{\mathbf{Cache}} = (\mathbf{Lab} \times \Delta) \rightarrow \widehat{\mathbf{Val}}\end{aligned}$$

- Values are now indexed by context
- This treatment prevents merging values from different calls (as in the previous example)

k -CFAs

- In **uniform k -CFA** we have a cache for each possible context

$$\hat{C} \in \widehat{\mathbf{Cache}} = (\mathbf{Lab} \times \Delta) \rightarrow \widehat{\mathbf{Val}}$$

where Δ : set of call strings (length $\leq k$)

- In the **classical k -CFA**, we have instead a cache for each possible full environment

more expensive

$$\hat{C} \in \widehat{\mathbf{Cache}} = (\mathbf{Lab} \times \mathbf{CEnv}) \rightarrow \widehat{\mathbf{Val}}$$

- In the **polynomial k -CFA**, instead of $\widehat{\mathbf{Val}} = \wp(\mathbf{Val} \times \mathbf{Cenv})$ we have

$$\hat{v} \in \widehat{\mathbf{Val}} = \wp(\mathbf{Val} \times \Delta)$$

To recap

$$ce \in \mathbf{Cenv} = \mathbf{Var} \rightarrow \Delta$$

$$\hat{\delta} \in \Delta = \mathbf{Lab}^{\leq k}$$

$$\hat{v} \in \widehat{\mathbf{Val}} = \wp(\mathbf{Val} \times \mathbf{Cenv})$$

$$\hat{\rho} \in \widehat{\mathbf{Cenv}} = (\mathbf{Var} \times \Delta) \rightarrow \widehat{\mathbf{Val}}$$

$$\hat{C} \in \widehat{\mathbf{Cache}} = (\mathbf{Lab} \times \Delta) \rightarrow \widehat{\mathbf{Val}}$$

Acceptability relation

$$(\hat{C}, \hat{\rho}) \models_{\delta}^{ce} e$$

- ce is the current context environment:
 - it tracks the context in which each variable of e is bound
 - it changes when variables are bound
- $\hat{\delta}$ is the current calling context
 - it changes when functions are applied

The formula expresses that $(\hat{C}, \hat{\rho})$ is an acceptable analysis of e in the context specified by ce and $\hat{\delta}$

Acceptability is now **context-dependent**: we analyse e in a specific calling context $\hat{\delta}$

Clause for variables

$$[var] (\hat{C}, \hat{\rho}) \models_{\delta}^{ce} x^{\ell} \text{ iff } \hat{\rho}(x, ce(x)) \subseteq \hat{C}(\ell, \delta)$$

We look up x in the context where it was bound and propagate it to the current context

Same variable,
different contexts $\rightarrow$ different values

- $ce(x)$ context where x was defined [definition context]
- δ : current context (where we use x) [use context]

A variable carries the context where it was created, but is used in the current context

Example

Suppose x was bound at call site 5, i.e., $ce_2(x) = 5$, then

$$(\hat{C}'_{id}, \hat{\rho}'_{id}) \models_5^{ce_2} x^1 \text{ iff } \hat{\rho}'_{id}(x, 5) \subseteq \hat{C}'_{id}(1, 5) \text{ [see Example 4]}$$

Clause for functions

$$[fn] (\hat{C}, \hat{\rho}) \models_{\delta}^{ce} (fn\ x \Rightarrow e_0)^{\ell} \text{ iff } \{(fn\ x \Rightarrow e_0, ce_0)\} \subseteq \hat{C}(\ell, \delta) \\ \text{where } ce_0 = ce \upharpoonright FV(fn\ x \Rightarrow e_0)$$

The functional term $fn\ x \Rightarrow e_0$ must be included in $\hat{C}(\ell, \delta)$ together with the current context environment ce , restricted to the free variables of the function (denoted ce_0)

Functions are represented as **closures**: a function is paired with the context in which it is defined, restricted to its free variables

Example: Suppose Λ be the empty sequence of labels and $ce = [x \mapsto 5, y \mapsto 3]$

$$(\hat{C}'_{id}, \hat{\rho}'_{id}) \models_{\Lambda}^{ce} (fn\ x \Rightarrow x^1)^2 \text{ iff } \{(fn\ x \Rightarrow x^1, ce_0)\} \subseteq \hat{C}'_{id}(2, \Lambda)$$

with $ce_0 = []$ since $FV(fn\ x \Rightarrow x^1) = \emptyset$ [see Example 4]

Note that if the function were $(fn\ x \Rightarrow y)$, then $FV(fn\ x \Rightarrow y) = \{y\}$

environment at application time

Clause for the let binding

$$[let] (\hat{C}, \hat{\rho}) \models_{\delta}^{ce} (\text{let } x = t_1^{\ell_1} \text{ in } t_2^{\ell_2})^{\ell} \text{ iff } (\hat{C}, \hat{\rho}) \models_{\delta}^{ce} t_1^{\ell_1} \wedge (\hat{C}, \hat{\rho}) \models_{\delta}^{ce'} t_2^{\ell_2} \wedge \hat{C}(\ell_1, \delta) \subseteq \hat{\rho}(x, \delta) \wedge \hat{C}(\ell_2, \delta) \subseteq \hat{C}(\ell, \delta) \\ \text{where } ce' = ce[x \mapsto \delta]$$

we record that x is defined in the current context δ

This clause

- requires checking the **sub-expressions**
- the definition context of the bound variable of the let-construct is the current context (as no application point is passed)

Double flow: propagation of the values

- from $\hat{C}(\ell_1, \delta)$ to $\hat{\rho}(x, \delta)$ [**binding**]
- from $\hat{C}(\ell_2, \delta)$ to $\hat{C}(\ell, \delta)$ [**evaluation**]

Example

$$(\hat{C}'_{id}, \hat{\rho}'_{id}) \models_{\Lambda}^{\square} (\text{let } f = (\text{fn } x \Rightarrow x^1)^2 \text{ in } (f^3 3^4)^5)^6 \text{ iff } (\hat{C}, \hat{\rho}) \models_{\Lambda}^{\square} t_1^{\ell_1} \wedge (\hat{C}, \hat{\rho}) \models_{\delta}^{[x \mapsto \delta]} t_2^{\ell_2} \wedge \hat{C}(2, \Lambda) \subseteq \hat{\rho}(f, \Lambda) \text{ and } \hat{C}(5, \Lambda) \subseteq \hat{C}(6, \Lambda)$$

Clause for function application

$$[app] (\hat{C}, \hat{\rho}) \models_{\delta}^{ce} (t_1^{\ell_1} t_2^{\ell_2})^{\ell} \text{ iff } (\hat{C}, \hat{\rho}) \models_{\delta}^{ce} t_1^{\ell_1} \wedge (\hat{C}, \hat{\rho}) \models_{\delta}^{ce} t_2^{\ell_2}$$

both subexpressions are analysed in the same context δ and environment ce

$$(\forall (\text{fn } x \Rightarrow t_0^{\ell_0}, ce_0) \in \hat{C}(\ell_1, \delta) :$$

$$(\hat{C}, \hat{\rho}) \models_{\delta_0}^{ce_0} t_0^{\ell_0} \wedge$$

operator may evaluate to a function closure

$$\hat{C}(\ell_2, \delta) \subseteq \hat{\rho}(x, \delta) \wedge \hat{C}(\ell_0, \delta) \subseteq \hat{C}(\ell, \delta))$$

body analysed in the updated context δ_0 using the closure environment

$$\text{where } \delta_0 = [\delta, \ell]_k \text{ and}$$

$$ce'_0 = ce_0[x \mapsto \delta_0]$$

we extend the current context with the call site ℓ

Propagation as before

x is now bound in the new calling context δ_0 but using the definition environment ce_0

Clause for function application

$$[app] (\hat{C}, \hat{\rho}) \models_{\delta}^{ce} (t_1^{\ell_1} t_2^{\ell_2})^{\ell} \text{ iff } (\hat{C}, \hat{\rho}) \models_{\delta}^{ce} t_1^{\ell_1} \wedge (\hat{C}, \hat{\rho}) \models_{\delta}^{ce} t_2^{\ell_2}$$

$\wedge \dots$

- We proceed in a similar way
- both f and x are bound in the new calling context δ_0 when analyzing the body

$$\begin{aligned} & (\forall (\text{fun } f \ x \Rightarrow t_0^{\ell_0}, ce_0) \in \hat{C}(\ell_1, \delta) : \\ & (\hat{C}, \hat{\rho}) \models_{\delta_0}^{ce'_0} t_0^{\ell_0} \wedge \\ & \hat{C}(\ell_2, \delta) \subseteq \hat{\rho}(x, \delta) \wedge \hat{C}(\ell_0, \delta) \subseteq \hat{C}(\ell, \delta)) \\ & \{(\text{fun } f \ x \Rightarrow t_0^{\ell_0}, ce_0)\} \in \hat{\rho}(f, \delta_0) : \\ & \text{where } \delta_0 = [\delta, \ell]_k \text{ and} \\ & ce'_0 = ce_0[f \mapsto \delta_0, x \mapsto \delta_0] \\ & \wedge \dots \end{aligned}$$

Uniform k -CFA

- [con] $(\widehat{C}, \widehat{\rho}) \models_{\delta}^{ce} c^{\ell}$ always
- [var] $(\widehat{C}, \widehat{\rho}) \models_{\delta}^{ce} x^{\ell}$ iff $\widehat{\rho}(x, ce(x)) \subseteq \widehat{C}(\ell, \delta)$
- [fn] $(\widehat{C}, \widehat{\rho}) \models_{\delta}^{ce} (\mathbf{fn} \ x \Rightarrow e_0)^{\ell}$ iff $\{(\mathbf{fn} \ x \Rightarrow e_0, ce_0)\} \subseteq \widehat{C}(\ell, \delta)$
where $ce_0 = ce \mid FV(\mathbf{fn} \ x \Rightarrow e_0)$
- [fun] $(\widehat{C}, \widehat{\rho}) \models_{\delta}^{ce} (\mathbf{fun} \ f \ x \Rightarrow e_0)^{\ell}$ iff $\{(\mathbf{fun} \ f \ x \Rightarrow e_0, ce_0)\} \subseteq \widehat{C}(\ell, \delta)$
where $ce_0 = ce \mid FV(\mathbf{fun} \ f \ x \Rightarrow e_0)$
- [app] $(\widehat{C}, \widehat{\rho}) \models_{\delta}^{ce} (t_1^{\ell_1} t_2^{\ell_2})^{\ell}$
iff $(\widehat{C}, \widehat{\rho}) \models_{\delta}^{ce} t_1^{\ell_1} \wedge (\widehat{C}, \widehat{\rho}) \models_{\delta}^{ce} t_2^{\ell_2} \wedge$
 $(\forall (\mathbf{fn} \ x \Rightarrow t_0^{\ell_0}, ce_0) \in \widehat{C}(\ell_1, \delta) :$
 $(\widehat{C}, \widehat{\rho}) \models_{\delta_0}^{ce'_0} t_0^{\ell_0} \wedge$
 $\widehat{C}(\ell_2, \delta) \subseteq \widehat{\rho}(x, \delta_0) \wedge \widehat{C}(\ell_0, \delta_0) \subseteq \widehat{C}(\ell, \delta)$
where $\delta_0 = [\delta, \ell]_k$
and $ce'_0 = ce_0[x \mapsto \delta_0]) \wedge$
 $(\forall (\mathbf{fun} \ f \ x \Rightarrow t_0^{\ell_0}, ce_0) \in \widehat{C}(\ell_1, \delta) :$
 $(\widehat{C}, \widehat{\rho}) \models_{\delta_0}^{ce'_0} t_0^{\ell_0} \wedge$
 $\widehat{C}(\ell_2, \delta) \subseteq \widehat{\rho}(x, \delta_0) \wedge \widehat{C}(\ell_0, \delta_0) \subseteq \widehat{C}(\ell, \delta) \wedge$
 $\{(\mathbf{fun} \ f \ x \Rightarrow t_0^{\ell_0}, ce_0)\} \subseteq \widehat{\rho}(f, \delta_0)$
where $\delta_0 = [\delta, \ell]_k$
and $ce'_0 = ce_0[f \mapsto \delta_0, x \mapsto \delta_0])$
- [if] $(\widehat{C}, \widehat{\rho}) \models_{\delta}^{ce} (\mathbf{if} \ t_0^{\ell_0} \mathbf{then} \ t_1^{\ell_1} \mathbf{else} \ t_2^{\ell_2})^{\ell}$
iff $(\widehat{C}, \widehat{\rho}) \models_{\delta}^{ce} t_0^{\ell_0} \wedge (\widehat{C}, \widehat{\rho}) \models_{\delta}^{ce} t_1^{\ell_1} \wedge (\widehat{C}, \widehat{\rho}) \models_{\delta}^{ce} t_2^{\ell_2} \wedge$
 $\widehat{C}(\ell_1, \delta) \subseteq \widehat{C}(\ell, \delta) \wedge \widehat{C}(\ell_2, \delta) \subseteq \widehat{C}(\ell, \delta)$
- [let] $(\widehat{C}, \widehat{\rho}) \models_{\delta}^{ce} (\mathbf{let} \ x = t_1^{\ell_1} \mathbf{in} \ t_2^{\ell_2})^{\ell}$
iff $(\widehat{C}, \widehat{\rho}) \models_{\delta}^{ce} t_1^{\ell_1} \wedge (\widehat{C}, \widehat{\rho}) \models_{\delta}^{ce'} t_2^{\ell_2} \wedge$
 $\widehat{C}(\ell_1, \delta) \subseteq \widehat{\rho}(x, \delta) \wedge \widehat{C}(\ell_2, \delta) \subseteq \widehat{C}(\ell, \delta)$
where $ce' = ce[x \mapsto \delta]$
- [op] $(\widehat{C}, \widehat{\rho}) \models_{\delta}^{ce} (t_1^{\ell_1} \ \mathbf{op} \ t_2^{\ell_2})^{\ell}$ iff $(\widehat{C}, \widehat{\rho}) \models_{\delta}^{ce} t_1^{\ell_1} \wedge (\widehat{C}, \widehat{\rho}) \models_{\delta}^{ce} t_2^{\ell_2}$

Back to Example 4: 0-CFA

$e \triangleq (\text{let } f = (\text{fn } x \Rightarrow x^1)^2$
 in $((f^3 f^4)^5 (\text{fn } y \Rightarrow y^6)^7)^8)^9$

The least 0-CFA analysis is given by $(\hat{C}_{id}, \hat{\rho}_{id})$

- $\hat{C}_{id}(1) = \{\text{fn } x \Rightarrow x^1, \text{fn } y \Rightarrow y^6\}$
- $\hat{C}_{id}(2) = \{\text{fn } x \Rightarrow x^1\}$
- $\hat{C}_{id}(3) = \{\text{fn } x \Rightarrow x^1\}$
- $\hat{C}_{id}(4) = \{\text{fn } x \Rightarrow x^1\}$
- $\hat{C}_{id}(5) = \{\text{fn } x \Rightarrow x^1, \text{fn } y \Rightarrow y^6\}$
- $\hat{C}_{id}(6) = \{\text{fn } y \Rightarrow y^6\}$
- $\hat{C}_{id}(7) = \{\text{fn } y \Rightarrow y^6\}$
- $\hat{C}_{id}(8) = \{\text{fn } x \Rightarrow x^1, \text{fn } y \Rightarrow y^6\}$
- $\hat{C}_{id}(9) = \{\text{fn } x \Rightarrow x^1, \text{fn } y \Rightarrow y^6\}$
- $\hat{\rho}_{id}(f) = \{\text{fn } x \Rightarrow x^1\}$
- $\hat{\rho}_{id}(x) = \{\text{fn } x \Rightarrow x^1, \text{fn } y \Rightarrow y^6\}$
- $\hat{\rho}_{id}(y) = \{\text{fn } y \Rightarrow y^6\}$

Only $\text{fn } y \Rightarrow y^6$ is a possible result of the evaluation of e
 0-CFA merges different calling contexts $\Rightarrow$ **loss of precision**

Back to Example 4: uniform 1-CFA

$e \triangleq (\text{let } f = (\text{fn } x \Rightarrow x^1)^2$
 $\text{in } ((f^3 f^4)^5 (\text{fn } y \Rightarrow y^6)^7)^8)^9$

- The initial context will be Λ , the empty sequence of labels
- In the course of the analysis the current context will be modified at the two application points with labels 5 and 8
- since we only records call strings of length at most 1 the only contexts of interest will therefore be in $\{\Lambda, 5, 8\}$
- In 0-CFA the two calls are merged
- In 1-CFA they are instead separated because they have different call sites: this is enough to restore precision

Back to Example 4: uniform 1-CFA

$e \triangleq (\text{let } f = (\text{fn } x \Rightarrow x^1)^2$
in $((f^3 f^4)^5 (\text{fn } y \Rightarrow y^6)^7)^8)^9$

- Context environments generated during the analysis:
 - $ce_0 = []$ initial (empty) context environment
 - $ce_1 = ce_0[f \mapsto \Lambda]$ used for analyzing the body of the let: f is defined in context Λ
 - $ce_2 = ce_0[x \mapsto 5]$ used for analyzing the body of f : x bound at call site 5
 - $ce_3 = ce_0[x \mapsto 8]$ used for analyzing the body of f : x bound at call site 8

We start from the closure environment ce_0 (captured at definition time), not from the current ce

Back to Example 4: uniform 1-CFA

$e \triangleq (\text{let } f = (\text{fn } x \Rightarrow x^1)^2$
 $\text{in } ((f^3 f^4)^5 (\text{fn } y \Rightarrow y^6)^7)^8)^9$

Look at the solution $(\hat{C}'_{id}, \hat{\rho}'_{id})$ • $\hat{C}'_{id}(1,5) = \{(\text{fn } x \Rightarrow x^1, ce_0)\}$ • $\hat{\rho}'_{id}(f, \Lambda) = \{(\text{fn } x \Rightarrow x^1, ce_0)\}$

• $\hat{C}'_{id}(2,\Lambda) = \{(\text{fn } x \Rightarrow x^1, ce_0)\}$ • $\hat{\rho}'_{id}(x,5) = \{(\text{fn } x \Rightarrow x^1, ce_0)\}$

• $\hat{C}'_{id}(4,\Lambda) = \{(\text{fn } x \Rightarrow x^1, ce_0)\}$ • $\hat{\rho}'_{id}(x,8) = \{(\text{fn } y \Rightarrow y^6, ce_0)\}$

With this CFA we can:

- distinguish the values of x in the two calling contexts
- determine that $\text{fn } y \Rightarrow y^6$ is the **only** result possible

• $\hat{C}'_{id}(7,\Lambda) = \{(\text{fn } y \Rightarrow y^6, ce_0)\}$

• $\hat{C}'_{id}(9,\Lambda) = \{(\text{fn } y \Rightarrow y^6, ce_0)\}$

• $\hat{C}'_{id}(1,8) = \{(\text{fn } y \Rightarrow y^6, ce_0)\}$

• $\hat{C}'_{id}(3,\Lambda) = \{(\text{fn } x \Rightarrow x^1, ce_0)\}$

• $\hat{C}'_{id}(5,\Lambda) = \{(\text{fn } x \Rightarrow x^1, ce_0)\}$

• $\hat{C}'_{id}(8,\Lambda) = \{(\text{fn } y \Rightarrow y^6, ce_0)\}$

Back to Example 4: 1-CFA

$e \triangleq (\text{let } f = (\text{fn } x \Rightarrow x^1)^2$
 $\quad \text{in } ((f^3 f^4)^5 (\text{fn } y \Rightarrow y^6)^7 (8)^9)$

We check acceptability

To check acceptability, i.e.,
 $(\hat{C}'_{id}, \hat{\rho}'_{id}) \models_{\Lambda}^{ce_0} e$ we proceed by
 clauses:

- [let], to check the update of the context environment
- [app] at label 8, to check check x^1 in context 8 and
- [app] at label 5, to check check x^1 in context 8
- for each closure, we check argument + body in updated context

- $\hat{C}'_{id}(1,5) = \{(\text{fn } x \Rightarrow x^1, ce_0)\}$
- $\hat{C}'_{id}(2,\Lambda) = \{(\text{fn } x \Rightarrow x^1, ce_0)\}$
- $\hat{C}'_{id}(4,\Lambda) = \{(\text{fn } x \Rightarrow x^1, ce_0)\}$
- $\hat{C}'_{id}(7,\Lambda) = \{(\text{fn } y \Rightarrow y^6, ce_0)\}$
- $\hat{C}'_{id}(9,\Lambda) = \{(\text{fn } y \Rightarrow y^6, ce_0)\}$
- $\hat{C}'_{id}(1,8) = \{(\text{fn } y \Rightarrow y^6, ce_0)\}$
- $\hat{C}'_{id}(3,\Lambda) = \{(\text{fn } x \Rightarrow x^1, ce_0)\}$
- $\hat{C}'_{id}(5,\Lambda) = \{(\text{fn } x \Rightarrow x^1, ce_0)\}$
- $\hat{C}'_{id}(8,\Lambda) = \{(\text{fn } y \Rightarrow y^6, ce_0)\}$
- $\hat{\rho}'_{id}(f, \Lambda) = \{(\text{fn } x \Rightarrow x^1, ce_0)\}$
- $\hat{\rho}'_{id}(x, 5) = \{(\text{fn } x \Rightarrow x^1, ce_0)\}$
- $\hat{\rho}'_{id}(x, 8) = \{(\text{fn } y \Rightarrow y^6, ce_0)\}$

Back to Example 4: 1-CFA

$e \triangleq (\text{let } f = (\text{fn } x \Rightarrow x^1)^2$
 $\text{in } ((f^3 f^4)^5 (\text{fn } y \Rightarrow y^6)^7)^8)^9$

We check acceptability

[let]: To check $(\hat{C}'_{id}, \hat{\rho}'_{id}) \models_{\Lambda}^{ce_0} e$ it suffices ([let]) to verify
 $(\hat{C}'_{id}, \hat{\rho}'_{id}) \models_{\Lambda}^{ce_0} (\text{fn } x \Rightarrow x^1)^2$ and $(\hat{C}'_{id}, \hat{\rho}'_{id}) \models_{\Lambda}^{ce_1} ((f^3 f^4)^5 (\text{fn } y \Rightarrow y^6)^7)^8$
 because $\hat{C}'_{id}(2, \Lambda) \subseteq \hat{\rho}'_{id}(f, \Lambda)$ and $\hat{C}'_{id}(8, \Lambda) \subseteq \hat{C}'_{id}(9, \Lambda)$

$$ce_0 = []$$

$$ce_1 = ce_0[f \mapsto \Lambda]$$

[app] at label 8: Since $\hat{C}'_{id}(5, \Lambda) = \{(\text{fn } x \Rightarrow x^1, ce_0)\}$, to check
 $(\hat{C}'_{id}, \hat{\rho}'_{id}) \models_{\Lambda}^{ce_1} ((f^3 f^4)^5 (\text{fn } y \Rightarrow y^6)^7)^8$ suffices ([app]) to verify
 $(\hat{C}'_{id}, \hat{\rho}'_{id}) \models_{\Lambda}^{ce_1} (f^3 f^4)^5$ and $(\hat{C}'_{id}, \hat{\rho}'_{id}) \models_{\Lambda}^{ce_1} (\text{fn } y \Rightarrow y^6)^7$ and $(\hat{C}'_{id}, \hat{\rho}'_{id}) \models_8^{ce_3} x^1$ iff $\hat{\rho}_{id}(x, 8) \subseteq \hat{C}'_{id}(8, \Lambda)$
 because $\hat{C}'_{id}(7, \Lambda) \subseteq \hat{\rho}'_{id}(x, 8)$ and $\hat{C}'_{id}(1, 8) \subseteq \hat{C}'_{id}(8, \Lambda)$

$$ce_3 = ce_0[x \mapsto 8]$$

$$\delta_0 = [\Lambda, 8]_1 = 8$$

[app] at label 5: Since $\hat{C}'_{id}(3, \Lambda) = \{(\text{fn } x \Rightarrow x^1, ce_0)\}$, to check
 $(\hat{C}'_{id}, \hat{\rho}'_{id}) \models_{\Lambda}^{ce_1} (f^3 f^4)^5$ suffices ([app]) to verify
 $(\hat{C}'_{id}, \hat{\rho}'_{id}) \models_{\Lambda}^{ce_1} f^3$ and $(\hat{C}'_{id}, \hat{\rho}'_{id}) \models_{\Lambda}^{ce_1} f^4$ and $(\hat{C}'_{id}, \hat{\rho}'_{id}) \models_5^{ce_2} x^1$
 because $\hat{C}'_{id}(4, \Lambda) \subseteq \hat{\rho}'_{id}(x, 5)$ and $\hat{C}'_{id}(1, 5) \subseteq \hat{C}'_{id}(5, \Lambda)$

$$ce_2 = ce_0[x \mapsto 5]$$

$$\delta_0 = [\Lambda, 5]_1 = 5$$

Back to Example 4: 1-CFA

$e \triangleq (\text{let } f = (\text{fn } x \Rightarrow x^1)^2$
 $\text{in } ((f^3 f^4)^5 (\text{fn } y \Rightarrow y^6)^7)^8)^9$

We check acceptability

[let]: To check $(\hat{C}'_{id}, \hat{\rho}'_{id}) \models_{\Lambda}^{ce_0} e$ it suffices ([let]) to verify
 $(\hat{C}'_{id}, \hat{\rho}'_{id}) \models_{\Lambda}^{ce_0} (\text{fn } x \Rightarrow x^1)^2$ and $(\hat{C}'_{id}, \hat{\rho}'_{id}) \models_{\Lambda}^{ce_1} ((f^3 f^4)^5 (\text{fn } y \Rightarrow y^6)^7)^8$
 because $\hat{C}'_{id}(2, \Lambda) \subseteq \hat{\rho}'_{id}(f, \Lambda)$ and $\hat{C}'_{id}(8, \Lambda) \subseteq \hat{C}'_{id}(9, \Lambda)$

$ce_0 = []$

$ce_1 = ce_0[f \mapsto \Lambda]$

[app] at label 8: Since $\hat{C}'_{id}(5, \Lambda) = \{(\text{fn } x \Rightarrow x^1, ce_0)\}$, to check
 $(\hat{C}'_{id}, \hat{\rho}'_{id}) \models_{\Lambda}^{ce_1} ((f^3 f^4)^5 (\text{fn } y \Rightarrow y^6)^7)^8$ suffices ([app]) to verify
 $(\hat{C}'_{id}, \hat{\rho}'_{id}) \models_{\Lambda}^{ce_1} (f^3 f^4)^5$ and $(\hat{C}'_{id}, \hat{\rho}'_{id}) \models_{\Lambda}^{ce_1} (\text{fn } y \Rightarrow y^6)^7$ and $(\hat{C}'_{id}, \hat{\rho}'_{id}) \models_8^{ce_3} x^1$ iff $\hat{\rho}_{id}(x, 8) \subseteq \hat{C}_{id}(8, \Lambda)$
 because $\hat{C}'_{id}(7, \Lambda) \subseteq \hat{\rho}'_{id}(x, 8)$ and $\hat{C}'_{id}(1, 8) \subseteq \hat{C}'_{id}(8, \Lambda)$

$ce_3 = ce_0[x \mapsto 8]$

[app] at label 5: Since $\hat{C}'_{id}(3, \Lambda) = \{(\text{fn } x \Rightarrow x^1, ce_0)\}$, to check
 $(\hat{C}'_{id}, \hat{\rho}'_{id}) \models_{\Lambda}^{ce_1} (f^3 f^4)^5$ suffices ([app]) to verify
 $(\hat{C}'_{id}, \hat{\rho}'_{id}) \models_{\Lambda}^{ce_1} f^3$ and $(\hat{C}'_{id}, \hat{\rho}'_{id}) \models_{\Lambda}^{ce_1} f^4$ and $(\hat{C}'_{id}, \hat{\rho}'_{id}) \models_5^{ce_2} x^1$
 because $\hat{C}'_{id}(4, \Lambda) \subseteq \hat{\rho}'_{id}(x, 5)$ and $\hat{C}'_{id}(1, 5) \subseteq \hat{C}'_{id}(5, \Lambda)$

$ce_2 = ce_0[x \mapsto 5]$

only $\text{fn } y \Rightarrow y^6$ can flow to $\hat{C}'_{id}(9, \Lambda)$

Correctness and implementation of the uniform k -CFA

Soundness

The analysis is sound w.r.t. the operational semantics

Implementation

Can be implemented via constraint solving / fixpoint iteration

Trade-off

- Higher $k \Rightarrow$ more precision
- But exponential worst-case complexity

Complexity of uniform k -CFA

- even for $k = 1$: exponential worst case complexity

Why? (intuition)

- Program size: n , number of variables: p
- Contexts:
 - Δ has $O(n)$ elements $\Rightarrow$ $O(n \cdot p)$ different pairs (x, δ) and $O(n^2)$ different pairs (ℓ, δ)
- Abstract states:
 - $(\hat{C}, \hat{\rho}) \approx O(n^2)$ of tuples of values from $\widehat{\mathbf{Val}}$ 
- Values:
 - $\widehat{\mathbf{Val}}$ itself is a powerset of pairs of the form $(t, ce) \Rightarrow O(n \cdot n^p)$ such pairs,
 - $\widehat{\mathbf{Val}}$ has height $O(n \cdot n^p)$ and, since $p = O(n)$, we have the exponential complexity

Complexity of 0-CFA

- Here for $k = 0$: exponential worst case complexity

Why? (intuition)

- Program size: n , number of variables: p
- No contexts:
 - Δ is a singleton $\Rightarrow$ $O(p)$ different pairs (x, δ) and $O(n)$ different pairs (ℓ, δ)
- Abstract states:
 - $(\hat{C}, \hat{\rho}) \approx O(p + n)$ of tuples of values from $\widehat{\mathbf{Val}}$
- Values:
 - $\widehat{\mathbf{Val}}$ (powerset of pairs of the form (t, ce)) $\Rightarrow$ but here only $(t, ce_0) \Rightarrow O(n)$
 - $\widehat{\mathbf{Val}}$ has height $O(n)$ and we have the polynomial complexity

Possible improvements of complexity

Problem:

- In k -CFA, $\widehat{\mathbf{Val}} = \wp(\mathbf{Val} \times \mathbf{Cenv})$
- Context environments cause exponential blow-up

Idea 1: reduce lattice height

- Using abstraction to limit $\widehat{\mathbf{Val}}$
- (Standard abstract interpretation techniques)

Possible improvements of complexity

Idea 2: simplify contexts $\widehat{\mathbf{Val}} = \wp(\mathbf{Val} \times \mathbf{Cenv})$

Replace context environments with contexts only

$$\widehat{\mathbf{Val}} = \wp(\mathbf{Term} \times \Delta)$$

closures = function +
full environment

closures = function +
call string

- No full environment
- Only call strings δ

Effect:

- No combinations of variables inside ce
- Number of abstract values drastically reduced
- Lattice of polynomial height
- Less precision, but much cheaper